

BARABINSK, Russian Federation

WMO: 296120

Lat: 55.3545N

Lon: 78.3078E

Elev: 392

StdP: 14.49

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-26.8	-37.8	0.6	-32.0	-32.6	0.9	-26.6	21.1	15.2	19.5	16.4	3.8	30	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	84.8	65.8	81.9	64.7	79.0	63.6	69.0	79.5	67.2	77.6	65.5	75.5	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.7	73.7	63.5	88.8	71.7	61.7	83.3	70.2	33.5	79.7	31.9	77.7	30.7	75.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
20.1	17.7	15.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-35.9	89.5	7.1	3.1	-41.0	91.7	-45.1	93.5	-49.1	95.2	-54.3	97.4
			-36.1	72.7	7.0	3.6	-41.1	75.3	-45.2	77.4	-49.2	79.4	-54.3	82.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	-1.8	3.8	18.7	38.3	53.0	62.9	66.0	61.2	50.1	37.5	18.6	5.3
	DBStd	26.05	14.61	13.02	10.95	10.45	9.93	7.82	5.92	6.74	8.16	8.91	14.04	14.68
	HDD50	7165	1606	1293	972	374	86	7	0	3	97	399	941	1387
	HDD65	11301	2071	1713	1437	802	388	127	61	154	451	854	1391	1852
	CDD50	1556	0	0	0	22	180	395	497	351	101	10	0	0
	CDD65	214	0	0	0	0	16	65	92	37	4	0	0	0
	CDH74	2077	0	0	0	3	226	635	788	375	50	0	0	0
	CDH80	470	0	0	0	0	48	153	186	76	7	0	0	0
Wind	WSAvg	7.4	7.2	7.5	8.1	8.4	8.2	6.8	6.3	6.3	6.9	7.4	8.1	8.1
Precipitation	PrecAvg	14.8	0.8	0.6	0.5	0.8	1.2	1.9	2.4	2.0	1.5	1.3	1.1	0.9
	PrecMax	22.2	2.0	1.1	2.0	2.4	2.9	6.1	5.8	4.1	3.4	2.6	2.3	2.4
	PrecMin	8.3	0.1	0.1	0.1	0.1	0.3	0.2	0.3	0.4	0.0	0.4	0.3	0.1
	PrecStd	2.8	0.4	0.3	0.4	0.5	0.7	1.3	1.5	0.9	0.9	0.6	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	33.4	42.5	72.4	84.9	87.8	88.9	86.8	79.3	66.3	45.2	34.2
		MCWB	30.8	31.6	36.7	54.3	60.7	66.6	69.4	66.0	60.4	51.3	42.0	32.3
	2%	DB	27.8	30.3	38.5	65.5	79.9	84.4	84.6	82.0	73.8	60.0	41.2	31.9
		MCWB	26.4	28.6	34.3	51.6	59.1	65.2	67.3	65.5	58.0	49.3	38.7	30.9
	5%	DB	24.0	26.7	35.8	59.8	75.7	81.0	81.5	78.1	68.4	54.7	37.9	28.7
		MCWB	22.8	25.3	32.8	49.0	57.2	64.1	66.1	63.3	55.9	46.4	35.7	27.5
	10%	DB	20.1	22.9	33.6	54.4	70.6	77.4	78.4	74.8	63.9	50.1	34.4	24.9
		MCWB	19.0	21.5	31.2	45.5	54.7	62.6	65.4	62.2	53.3	44.5	32.6	23.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	31.9	37.6	56.5	63.7	70.2	72.1	69.5	62.9	53.4	42.6	32.9
		MCDB	32.4	33.2	41.5	68.6	80.5	82.9	83.5	79.7	74.3	63.0	44.1	33.7
	2%	WB	26.6	28.7	35.0	52.8	60.7	67.3	69.8	67.2	59.3	50.4	38.8	30.8
		MCDB	27.8	30.3	37.8	63.5	76.9	79.2	79.8	78.1	71.1	57.2	40.9	32.0
	5%	WB	22.9	25.3	33.2	49.4	58.4	65.6	68.2	65.3	56.9	47.6	35.7	27.4
		MCDB	23.9	26.7	35.6	59.1	72.7	77.2	78.3	75.1	66.3	54.1	37.7	28.6
	10%	WB	19.0	21.5	31.2	46.1	55.9	64.0	66.5	63.3	54.5	44.5	32.9	23.7
		MCDB	20.1	22.8	33.6	54.2	68.0	75.0	75.9	72.4	62.8	49.5	34.2	24.8
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.9	15.4	16.4	19.8	18.6	18.1	18.8	17.5	13.3	11.1	13.1
		MCDBR	14.9	13.0	14.1	24.0	26.2	24.1	23.0	24.6	24.8	21.6	10.0	13.3
		MCWBR	14.9	12.6	11.5	13.7	11.0	9.7	9.2	10.5	12.0	12.4	8.6	12.8
	5% WB	MCDBR	15.0	13.0	12.3	23.0	23.7	21.0	20.2	20.5	22.6	19.5	9.9	13.5
		MCWBR	15.0	12.7	10.5	13.7	11.4	9.6	9.5	10.0	11.7	12.3	8.9	13.3
Clear-Sky Solar Irradiance	taub	0.286	0.317	0.373	0.408	0.402	0.408	0.434	0.416	0.365	0.351	0.319	0.288	
	taud	2.024	1.992	1.921	2.099	2.228	2.285	2.207	2.233	2.333	2.231	2.075	1.973	
	Ebn at Noon	188	228	242	256	266	266	255	252	250	218	172	152	
	Edh at Noon	22	33	45	44	41	39	42	38	30	26	21	18	
All-Sky Solar Radiation	RadAvg	294	628	1106	1463	1729	1880	1710	1392	969	483	229	187	
	RadStd	24	35	58	114	149	175	113	123	95	53	41	22	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page